Basic Principles of Medicine 2
Module : Digestion and Metabolism
Lecture No: 14

Title: Metabolism of Fructose and Galactose

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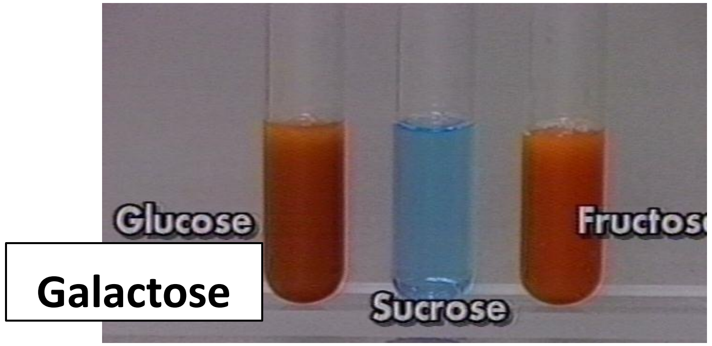
Reducing Sugars in Urine

- An 8-month-old child has **presence** of **reducing sugar in urine**. Urine sample is **positive** for **dipstick** test with **Clinitest**.
  - **Reducing** sugars **convert cupric** ions **to cuprous** ions
- What disorders do you think of in this clinical situation?
- What other clinical signs and symptoms are present?



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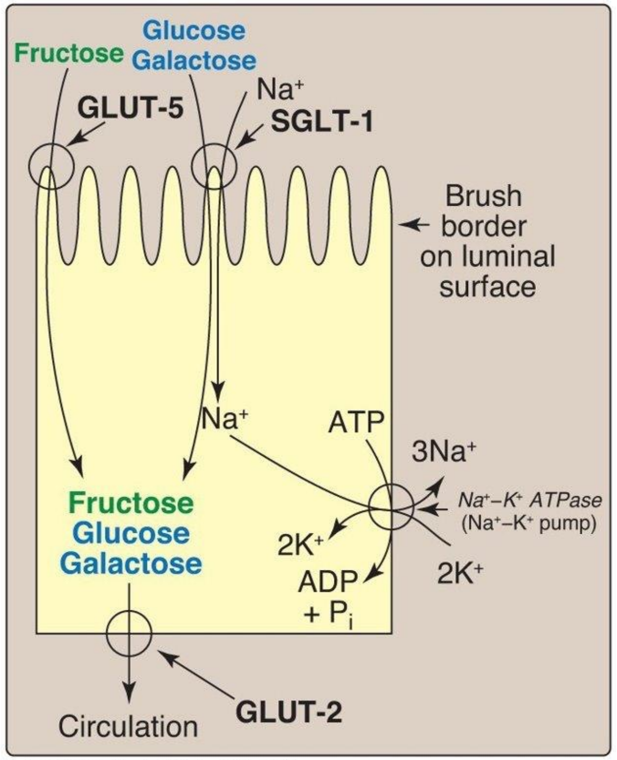
Fructose Metabolism



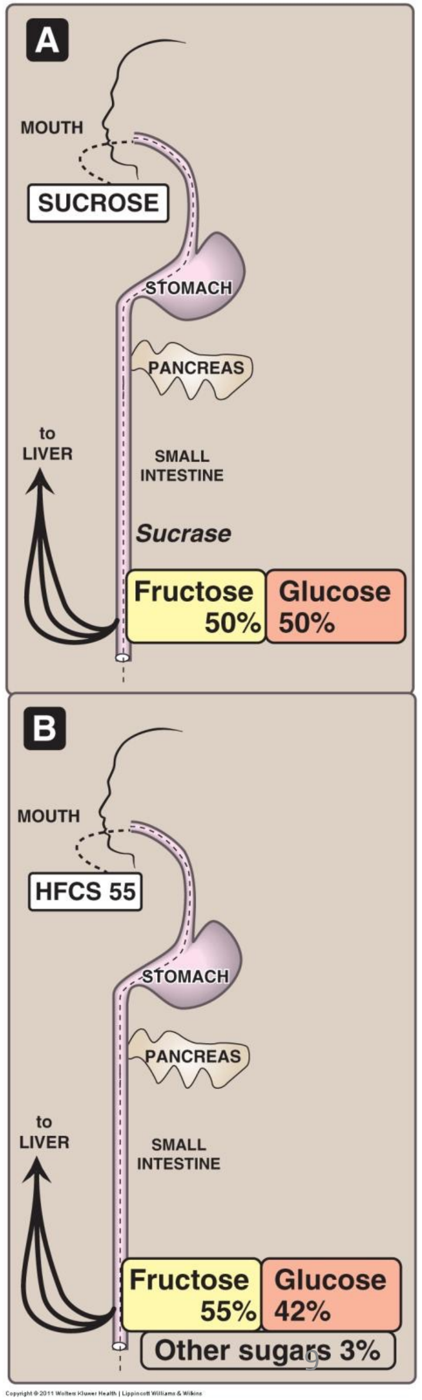
<u>**Dietary** sources of **Fructose**</u>

- **Sucrose,** digested by sucrase
- **Fruits** and **Honey**
- **High Fructose Corn Syrup** (55% fructose/ 45% glucose)
- **Sorbitol** = (**forms Fructose** by **Sorbitol Dehydrogenase**)

- **Fructose** is **absorbed by GLUT-5** on **luminal** side of intestinal cells
- **Metabolized in Liver**
- **Fructokinase**
- **Aldolase B** in **Liver** (same Aldolase B in Glycolysis), **has** a **higher affinity for F16BP** (Glycolysis) **than** for **F1P** (Fructose metabolism)

High Fructose Corn Syrup (HFCS)

- **Food/Beverage Sweetener**
- Higher incidence of **obesity**
- **Limit** dietary **HFCS** content
- **Processed foods** and **beverages**



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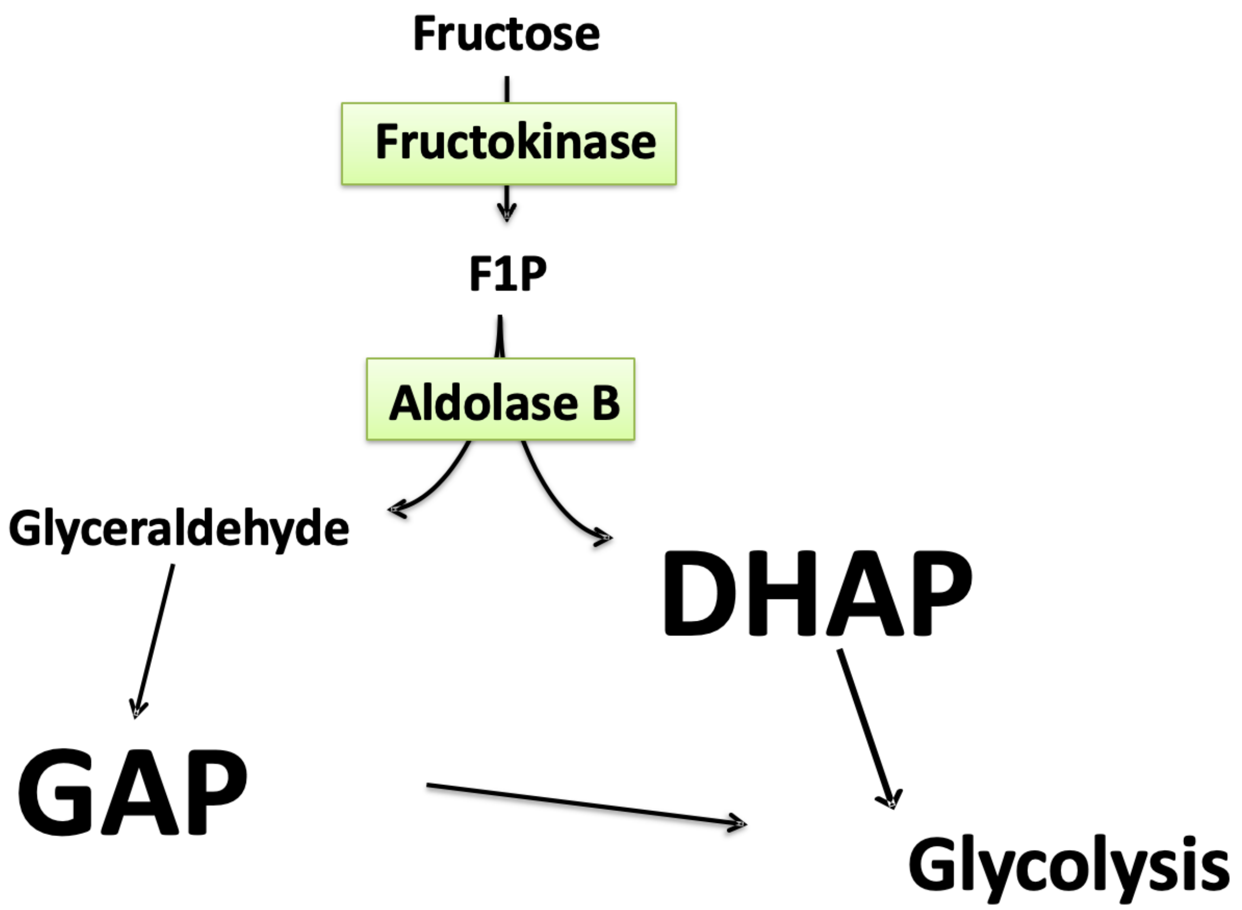
Fructose Metabolism

Aldolase B has a **higher affinity for F16BP than** for **F1P**

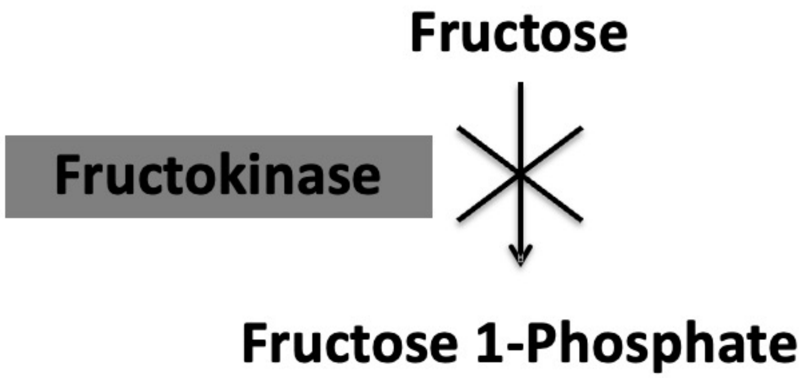
Fructose metabolism is **hindered BECAUSE Aldolase B prefers Glycolysis** & **Gluconeogenesis**



Case Report

- An 18-year-old girl avoids eating fruits and table sugar
- **Severe weakness** and **hypoglycemia** (Remember: hypoglycemia a few hours after sucrose or fructose, **trigger**!!!)
- **Drowsy** and **apathetic** (symptoms of hypoglycemia)
- Mom eliminated these foods
- Elimination of foods (trigger), symptoms less frequent
- Name foods eliminated by Candice’s mom
- Is this Essential fructosuria or Hereditary fructose intolerance?

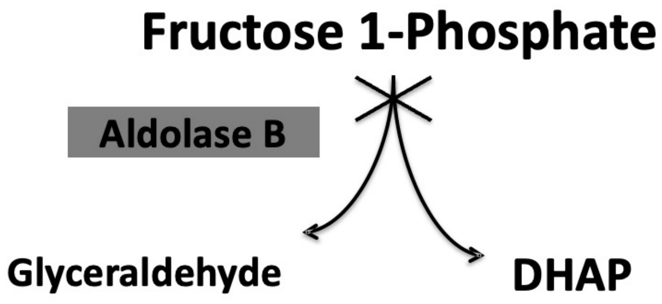
Benign Fructosuria (Essential Fructosuria)

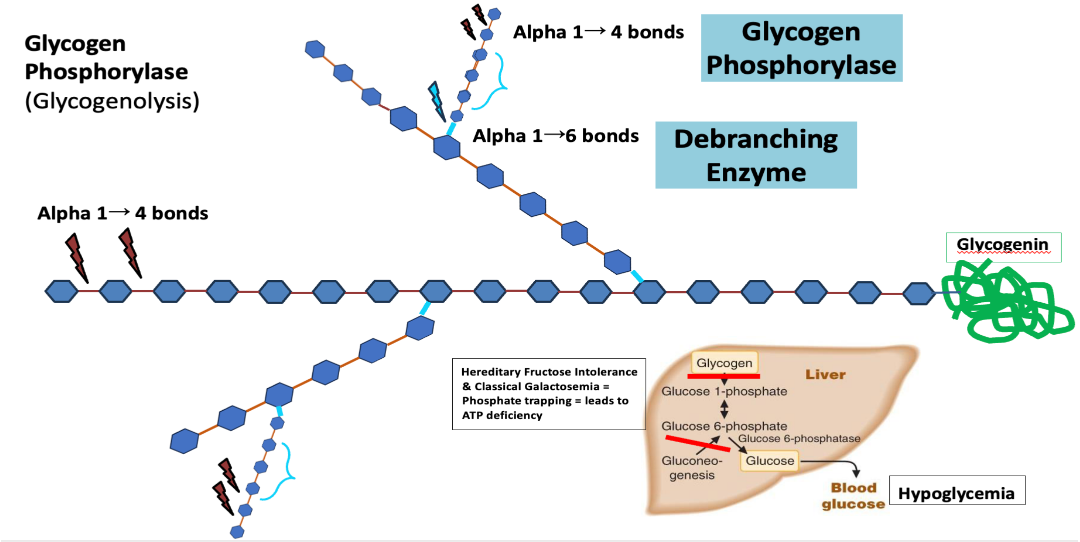


- Fructosuria = Liver Fructokinase deficiency

- **Fructose NOT metabolized** and **excreted in urine**
- **No toxic metabolites** accumulate, therefore **Asymptomatic**
- **Urinalysis shows Reducing sugar**, that is NOT glucose or galactose
- Detected by abnormal urinalysis report

Hereditary Fructose Intolerance





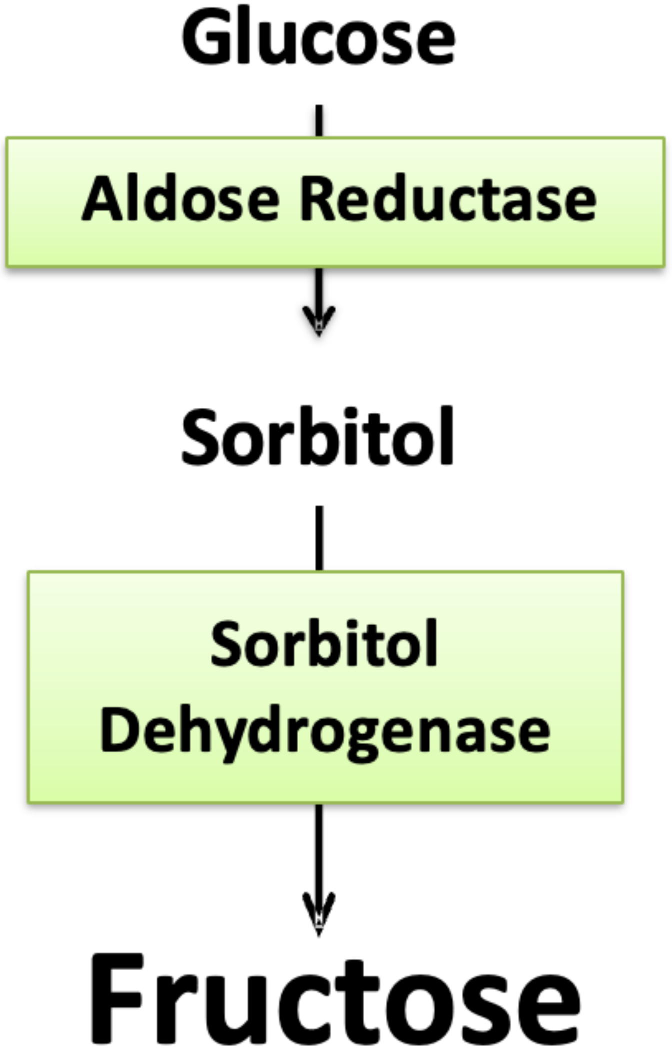
- **Hereditary Fructose Intolerance = Liver Aldolase B deficiency**
- Dietary **Sucrose/Fructose consumption + inability to metabolize beyond Aldolase B = F1P accumulates**
- **TRAPPING** of **inorganic phosphate (Pi),** results in **ATP deficiency = inhibits Gluconeogenesis & Glycogenolysis**

(Glycogen Phosphorylase requires inorganic phosphate)

- **Hypoglycemia** (**drowsy** and **apathetic**) **FOLLOWING** consumption of **Sucrose or Fructose** (**trigger**)
- **Symptoms after weaning** and **fruits added** (**6-8 months of life**)
- **Continued** dietary **Sucrose/Fructose/Sorbitol** = **Hepatocellular failure** and **Jaundice** (**Elevated AST & ALT**)
- **Urinalysis** shows **Reducing Sugar** (Fructose) that is NOT glucose
- Children with **Fructose intolerance DO NOT HAVE cataracts**
- **Avoid** dietary **Fructose** = **good dental** health and **NO dental caries**
- **Have aversion to sweets**
- **Diagnosis = Hypoglycemia FOLLOWING Sucrose/Fructose + Reducing Sugar in urine + Enzyme Assays**

Polyol Pathway = Synthesis of Fructose FROM Glucose

- **Fructose** can be synthesized **from Sorbitol**
  - (Those with **Fructose intolerance = MUST avoid Sorbitol because Sorbitol can become Fructose**)
- **Sorbitol** (sugar alcohol) = **from Glucose** by Aldose Reductase
- **Aldose Reductase** is found **rich in lens**, **retina**, **peripheral nerves** and **seminal vesicles**
- **Seminal Vesicles** are **rich in Sorbitol DH**, therefore the **Polyol Pathway** always ***occurs* here = *always* producing Fructose** (Fructose = main fuel for spermatozoa)



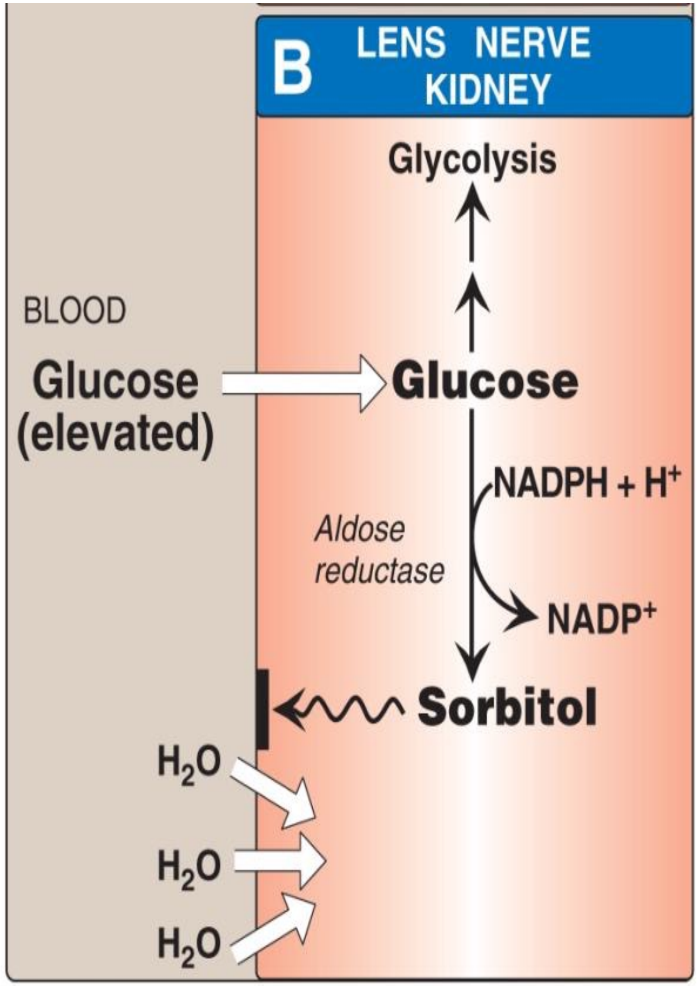
Significance of Polyol Pathway

<u>**In Uncontrolled Diabetes Mellitus,**</u>

- The **excess** of **plasma Glucose enters lens** and **forms Sorbitol**

- **Sorbitol** is an **osmotically** active molecule = **increases water** content **in lens**, results in **Cataracts (decreased lens transparency/increased lens opacity)**

- **Sorbitol** is **partly implicated in** the **pathogenesis of Microvascular complications** of **Diabetes Mellitus** (**neuropathy**, **nephropathy**, **retinopathy**) due to **Insulin-independent Glucose uptake**



Significance of Polyol Pathway

<u>Polyol Pathway in Galactosemia</u>

- **Elevated** blood **Galactose** = **entry into lens**

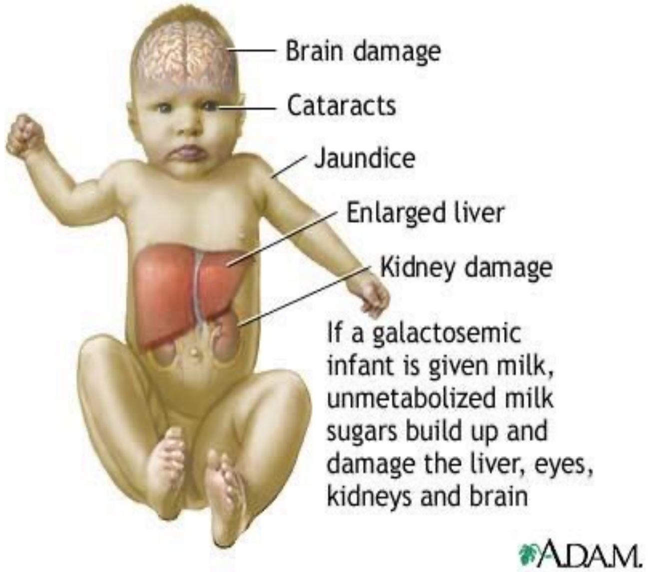
- **Galactose ----------Aldose Reductase---------> Galactitol**

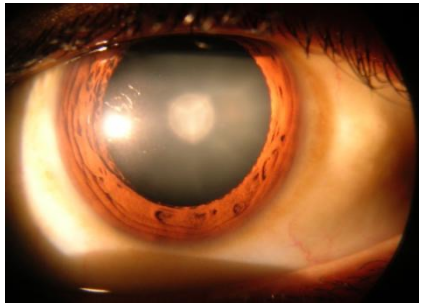
- **Galactitol** is an **osmotically active** molecule = **increases water** content **in lens =** results in **Cataracts (decreased lens transparency/increased lens opacity)**

- **Infants** with **untreated Galactosemia** have **Cataracts**

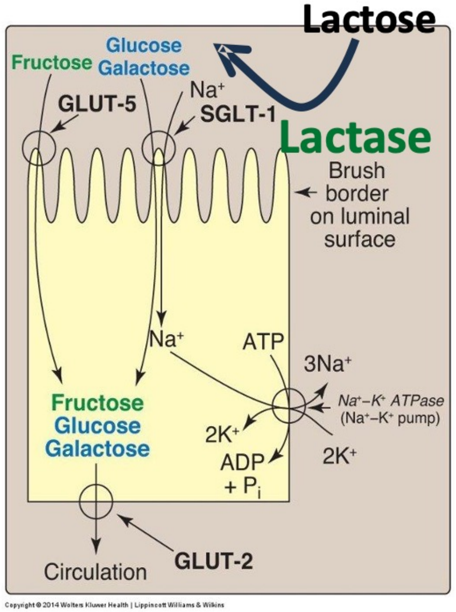
Case Report on Galactosemia

- A **3-week-old** infant has **worsening jaundice**
- **Vomiting after feeding**
- **Failure to thrive**
- **Higher risk of sepsis (E.coli)**
- **Hepatomegaly**
- Early **Cataract** formation
- **Developmental milestone delay**
- **Classical triad = Liver damage + Developmental delay + Cataracts**
- **Urine positive** for **Reducing Sugar** (not glucose)
- **Inborn error** of **Galactose** metabolism?





Galactose Metabolism



- **Lactose** (**milk** sugar) (**β1→4** linkage)

- **Intestinal Lactase** cleaves **to Galactose + Glucose**

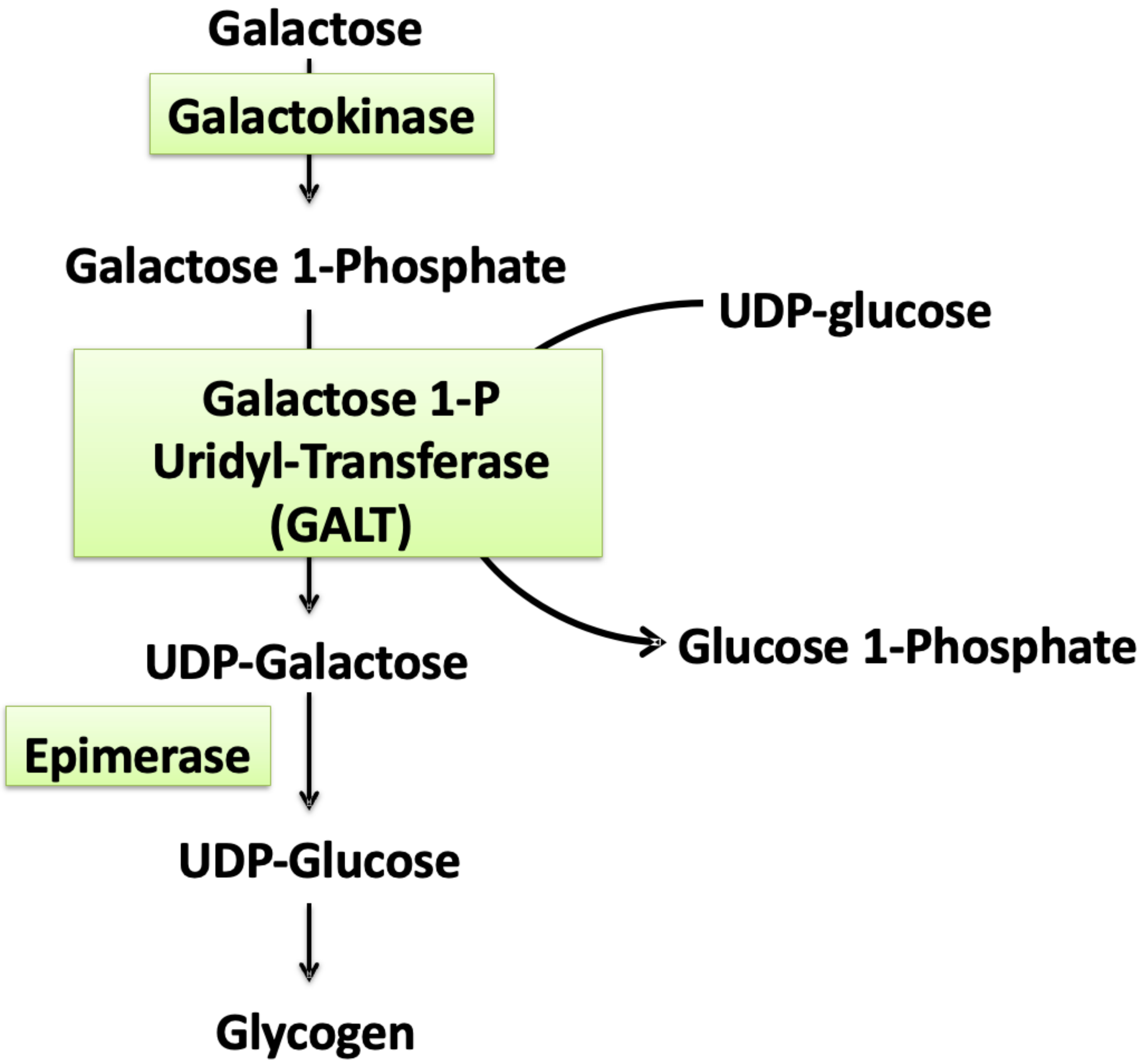
- **SGLT-1 imports Galactose** on **luminal side** of intestinal cells

- Metabolized in Liver

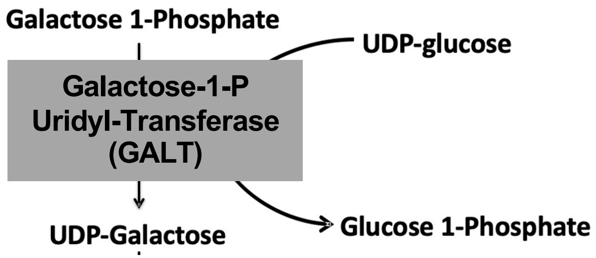
- **Galactokinase**, **Galactose 1-phosphate Uridyl Transferase (GALT enz.)** and **Epimerase**
-
- **Galactose eventually** converted **to Glucose** → **Glycogen**

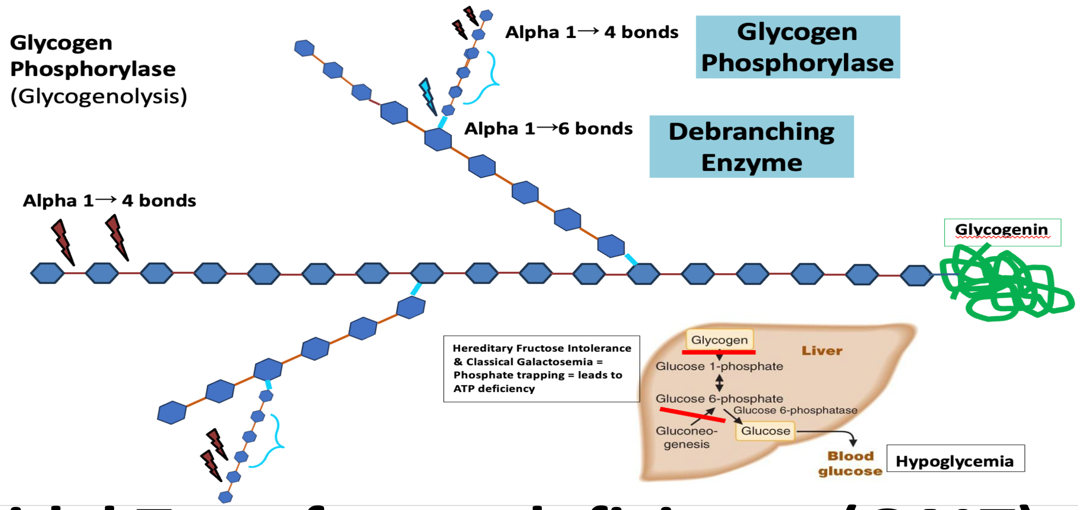
- **Lactose intolerance = Deficiency** of **intestinal lactase** = **Diarrhea, bloating, cramps are symptoms of lactose's osmtic influence in the lumen,** tests show **No Reducing Sugar** in **urine**; **No Liver involvement/Cataracts/developmental delay**

Galactose Metabolism

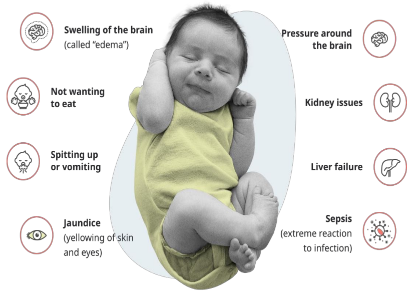


Classical Galactosemia



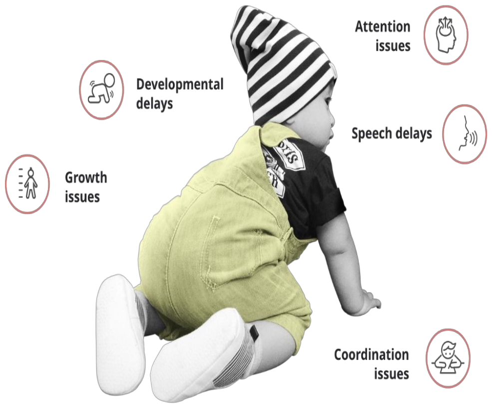


- **GALACTOSEMIA** = **Galactose 1-Phosphate Uridyl Transferase deficiency (GALT)**
- **Autosomal recessive**
- **Galactose** & **Galactose 1-P ACCUMULATE**
- Presentation: **2nd -3rd week of life**
- Higher risk of **SEPSIS** in **neonates**
- **Galactose 1-P** contributes to **Phosphate Trapping = decreased Gluconeogenesis** & **Glycogenolysis** = leads to **Hypoglycemia**
- **Hypoglycemia** (**drowsy** and **apathetic**) **following** dietary Lactose (milk) or Galactose **consumption** (**trigger**)
- **Gal-1-P** & **Galactitol ACCUMULATION = Hepatomegaly + Jaundice + Elevated AST** and **ALT** levels
- **Lens** = Galactose → **Galactitol** = **Cataracts**
- **Gal-1-P & Galactitol ACCUMULATE in brain** = **neurological damage, learning disability, milestone delay**
- **Newborn screening** = **Heel-prick test**
- **Measure Galactose OR Gal-1- P OR GALT activity**
- **Urinalysis = Reducing Sugar (Galactose)** that is NOT glucose



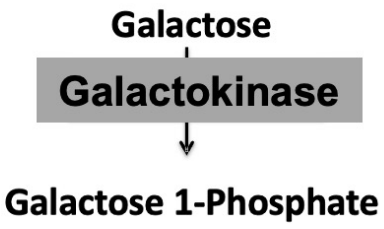
Management of Galactosemia

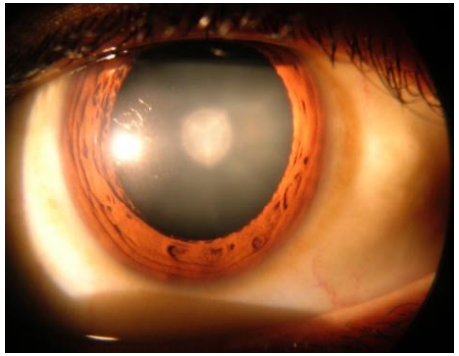
- **Dietary EXCLUSION of** = **Galactose / Lactose**
  - **Improves** = **Liver function** and **Cataracts**
  - **Limits** the effect on **learning disability,** varied



- **Breast-feeding** must be **STOPPED**
- **SUBSTITUTE** with **Soy-milk OR Lactose-Free milk (not Lactaid)**
- **LIFELONG dietary restrictions**
- **MONITOR Gal-1-P levels**
- **Females WITH treated Galactosemia**, have **delayed puberty** and **premature ovarian failure**
- **Many treated children** also **have developmental delays** and **speech problems**

Non-classical Galactosemia





- **NON-classical GALACTOSEMIA** = **Galactokinase deficiency**
- **Lens Galactitol** formation → **Cataracts**
- **NO Gal 1-P accumulation = No Phosphate Trapping = No Liver damage**
- **Less severe** than Classical Galactosemia
- **Urine positive** for **Reducing Sugar (Galactose)**
- **Monitor Galactitol** levels
- **Dietary Galactose / Lactose restriction**

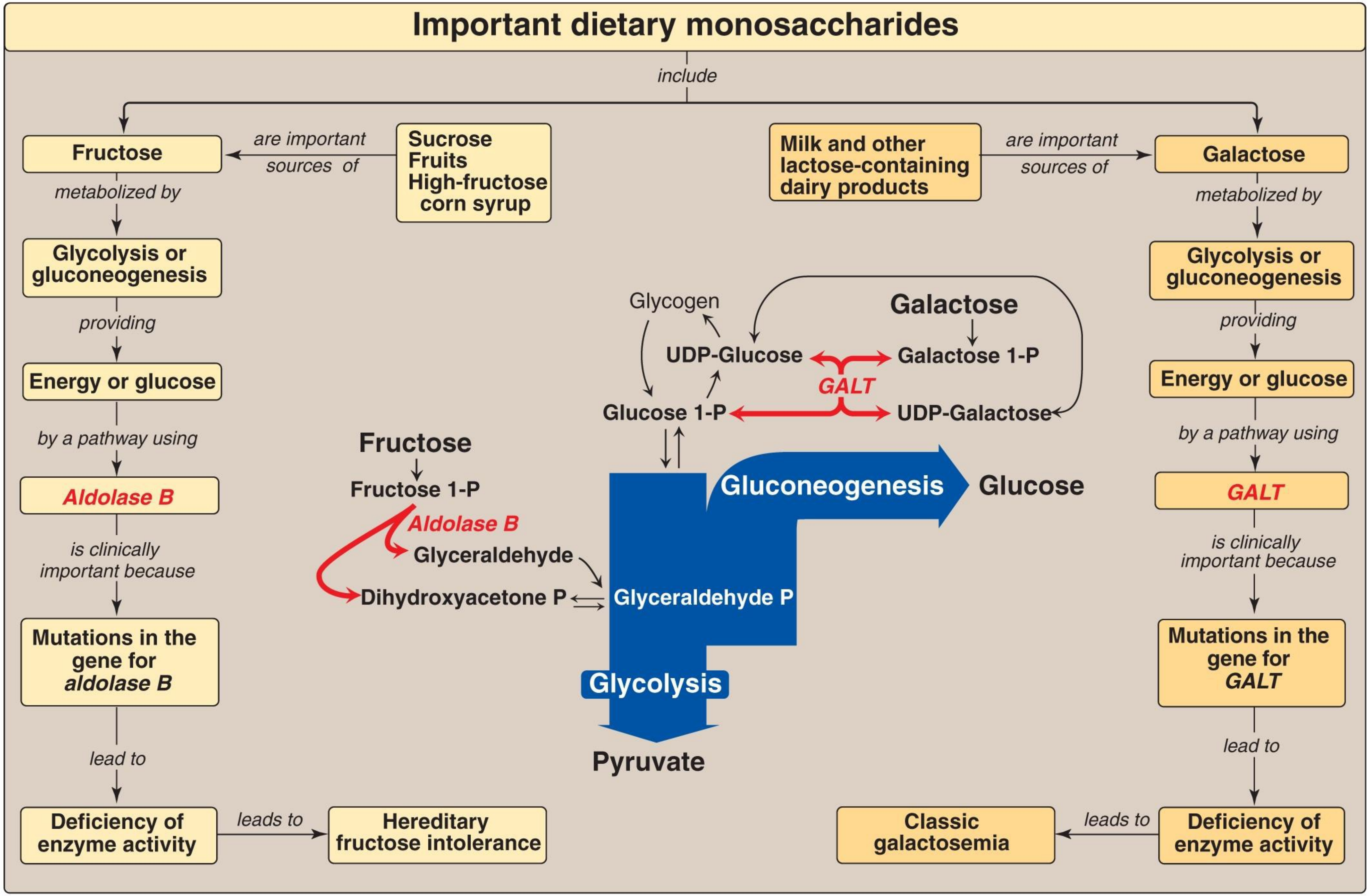
Reducing Sugar in Urine - Differential Diagnosis

An **8-month-old** child has a **positive** dipstick test with **Clinitest. What disorders** do you think of in this clinical situation?
Inherited disorders of Fructose or Galactose metabolism? What other clinical signs and symptoms are present?

- **Cataracts + Liver** disease + **Developmental** delay **= Classical Galactosemia**
- **Cataracts ONLY = Non-classical Galactosemia**
- **Hypoglycemia** + **Liver disease = Classical Galactosemia OR Hereditary Fructose Intolerance**
- **Hypoglycemia AFTER Sucrose** ingestiion**= Hereditary Fructose Intolerance)**
- **Hypoglycemia AFTER Lactose** ingestion **= Classical Galactosemia = GALT deficiency**
- **Benign** (**No significant** clinical **features**) = **(Benign Fructosuria)**

Lactose Intolerance

	Enzyme deficient and accumulated substrate	Predominant clinical features	Biochemical basis for clinical features	Biochemical basis for management
Lactose intolerance	**Intestinal Lactase** (disaccharidase)	**diarrhea**, **bloating** (**No Reducing** sugar in urine; **No Cataracts**, **No liver damage**, or **No Developmental delay)**	**Intestinal undigested lactose** is **degraded by bacteria** into **gas**	**Lactose avoidance** in diet



Lippincott's fig

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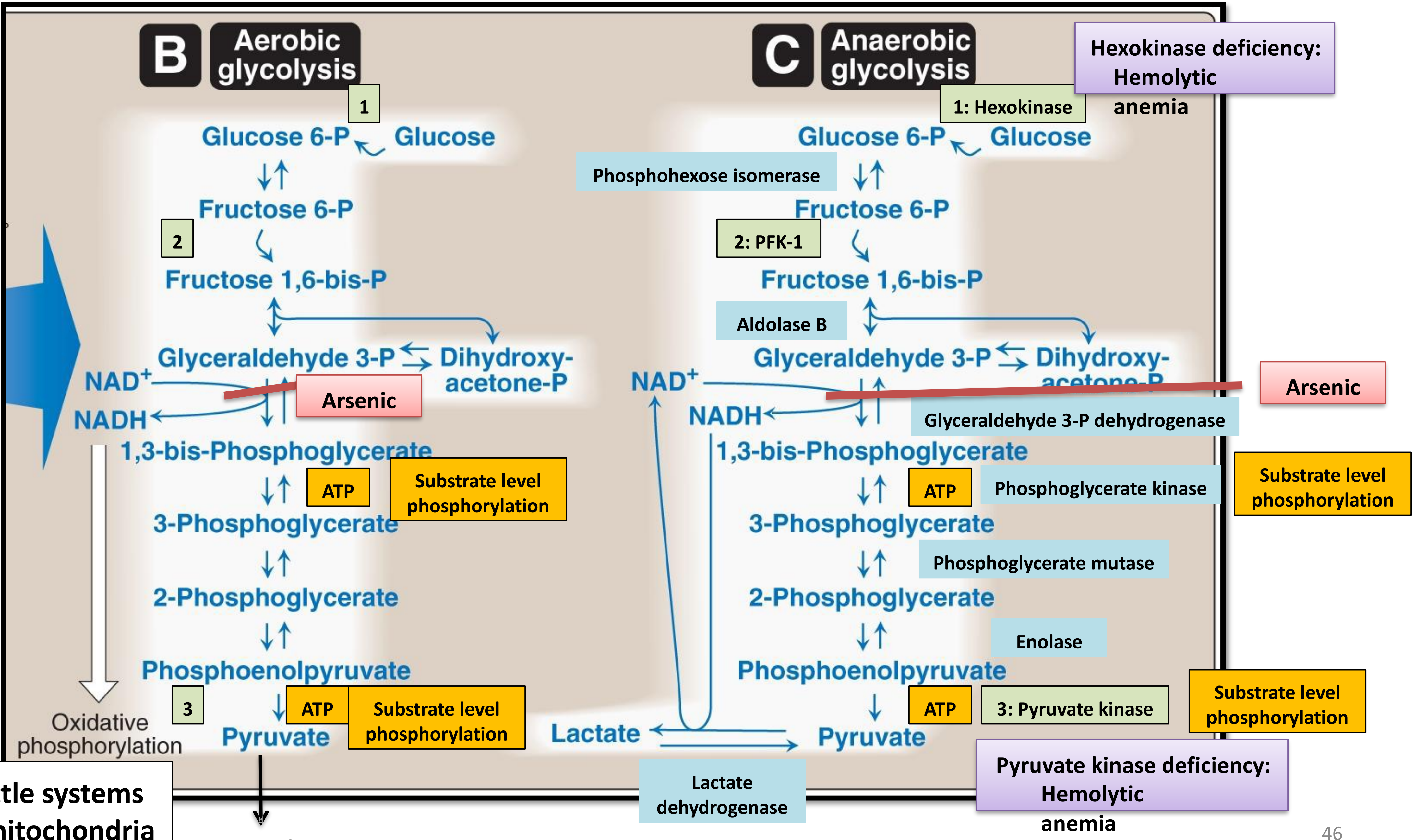

Via Shuttle systems enters mitochondria

Acetyl CoA

L. pg. 96, Fig 8.9